

AP CALCULUS AB - UNIT 1

Guided Notes 1.1: Can Change Occur at an Instant?



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we interpret average rate of change in a context, including correct units?

- Interpret average rate of change in a context, including correct units.
- Use shrinking intervals to estimate a rate at one instant.
- Connect secant slopes, tangent slopes, and limit notation.
- Explain why an instantaneous rate is defined through nearby values rather than a zero-length interval.

Key Knowledge, Vocabulary, and Notation

Average rate of change $\frac{f(b) - f(a)}{b - a}$ measures output change per unit input on an interval.

Instantaneous rate $\lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}$, when this limit exists.

Geometric meaning Secant slopes approach the slope of the tangent line.

Context and units The units are always output units divided by input units.

Zero interval warning $\frac{f(c) - f(c)}{c - c}$ is undefined; a limit studies nonzero intervals that shrink toward zero.

Concept Development and Strategy

1. Compute an average rate first. Then shorten the interval while keeping the target instant fixed.
2. For a left estimate use inputs below the target; for a right estimate use inputs above the target. Agreement supports a two-sided instantaneous rate.
3. A numerical estimate should include units and a contextual sentence. A symbolic answer should display the difference quotient before simplification.

Worked Example: Polynomial motion

A particle has position $s(t) = t^3 - 2t$ meters. Estimate its instantaneous velocity at $t = 2$ using $[2, 2.01]$ and $[1.99, 2]$.

Answer and Reasoning

$s(2) = 4$. The right secant slope is $\frac{s(2.01) - s(2)}{0.01} = 10.0601$ and the left secant slope is $\frac{s(2) - s(1.99)}{0.01} = 9.9401$. Both approach 10, so the instantaneous velocity is approximately 10 m/s. In exact limit form, $\lim_{h \rightarrow 0} \frac{(2+h)^3 - 2(2+h) - 4}{h} = 10$.

Worked Example: Interpreting units

The amount of water in a tank is $V(t)$ liters, where t is measured in minutes. Explain the meaning of $\frac{V(18)-V(12)}{6} = -3.4$.

Answer and Reasoning

From minute 12 to minute 18, the tank loses water at an average rate of 3.4 liters per minute. The negative sign records a decrease.

Worked Example: A radical rate

Evaluate $\lim_{h \rightarrow 0} \frac{\sqrt{9+h}-3}{h}$ and interpret it as the instantaneous rate of $f(x) = \sqrt{x}$ at $x = 9$.

Answer and Reasoning

Multiply by the conjugate: $\frac{\sqrt{9+h}-3}{h} \cdot \frac{\sqrt{9+h}+3}{\sqrt{9+h}+3} = \frac{1}{\sqrt{9+h}+3}$. The limit is $\frac{1}{6}$, so the tangent slope at $x = 9$ is $\frac{1}{6}$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. For $s(t) = t^2 + 3t$, find the average rate of change on $[1, 4]$.

2. For $q(t) = 120 - 5t + 2t^2$, find the average rate of change on $[2, 5]$.

3. For $s(t) = t^3 - 2t$, calculate the secant slope on $[2, 2.1]$.

4. For the same function, calculate the secant slope on $[2, 2.01]$ and compare it with the slope on $[1.99, 2]$.

5. A fish population is modeled by $P(t) = 1000 + 40t - 3t^2$. Find the instantaneous rate of change at $t = 4$ using a limit or an equivalent algebraic derivation.

6. For $f(x) = ax^2 + bx + d$, simplify $\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$.

7. Construct a quadratic position function $s(t) = at^2 + bt + c$ satisfying $s(0) = 2$, $s(1) = 5$, and instantaneous velocity 7 at $t = 1$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. For $f(x) = x^2 - 4x$, which expression represents the instantaneous rate at $x = 3$?

- (A) $\frac{f(3)-f(3)}{3-3}$ (B) $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$
 (C) $\lim_{x \rightarrow 0} \frac{f(x) - f(3)}{x - 3}$ (D) $\frac{f(4)-f(2)}{2}$

2. A particle has $s(t) = 5t^2 - t$. Its instantaneous velocity at $t = 2$ is

- (A) 9 (B) 10
 (C) 19 (D) 20

3. For $P(t) = \frac{800}{1+3e^{-0.4t}}$, a calculator gives $P(4.99) \approx 568.330$ and $P(5.01) \approx 569.645$. Which is the best estimate of $P'(5)$?

- (A) 0.658 (B) 6.58
 (C) 65.8 (D) 131.5

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Suppose every secant slope from $(c, f(c))$ to $(c + h, f(c + h))$ equals $m + kh$ for nonzero h near 0. Prove that the instantaneous rate at c exists and find it.

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